AI - 101

Breast Cancer
Logistic Regression

REPO LINK

https://github.com/AnadiKrishnAgnihotri/breast-cancer-logistic-regression.git

THE TEAM

PPT Lead

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Problem and Data

1 What and Why

What?

- Breast cancer is one of the most common cancers worldwide.
- Early identification of breast cancer can help improve treatment outcomes.
- Our project uses Logistic Regression, a machine learning algorithm, to analyse breast cancer data and predict a numerical target based on tumour-related features.

Why?

- To understand patterns in breast cancer data.
- To identify relationships between tumour characteristics and the target variable.
- To build and evaluate a machine learning model using Python.
- To explore how data-driven predictions can support medical research.

2 Data Set Source

We use the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, available through the UCI Machine Learning Repository.

The dataset is included in this repository as

`data/data.csv`



3 Size, Columns

569

rows representing breast tumour samples

30

numerical features describing cell nuclei characteristics

Target column: Diagnosis

Classes: Malignant (M) and Benign (B)

Approach

1 Pipeline Diagram

Dataset Collection
Breast Cancer Wisconsin Dataset

Data Preprocessing
Cleaning and feature selection

Train-Test Split
Training data and testing data

Logistic Regression Model
Model training

Prediction
Generate predicted values

Model Evaluation
Compare actual and predicted values

2 Digital Complexity

Data	Input tumour characteristics such as radius, texture, perimeter and area.
Model	Logistic Regression analyses relationships between input features and the target.
Output	Generates a predicted numerical value based on the trained model.

3 Libraries Used

Pandas
Data manipulation and analysis

NumPy
Numerical computations

Matplotlib
Data visualization

Scikit-learn
Machine learning model development and evaluation

Results

Testing set evaluation metrics

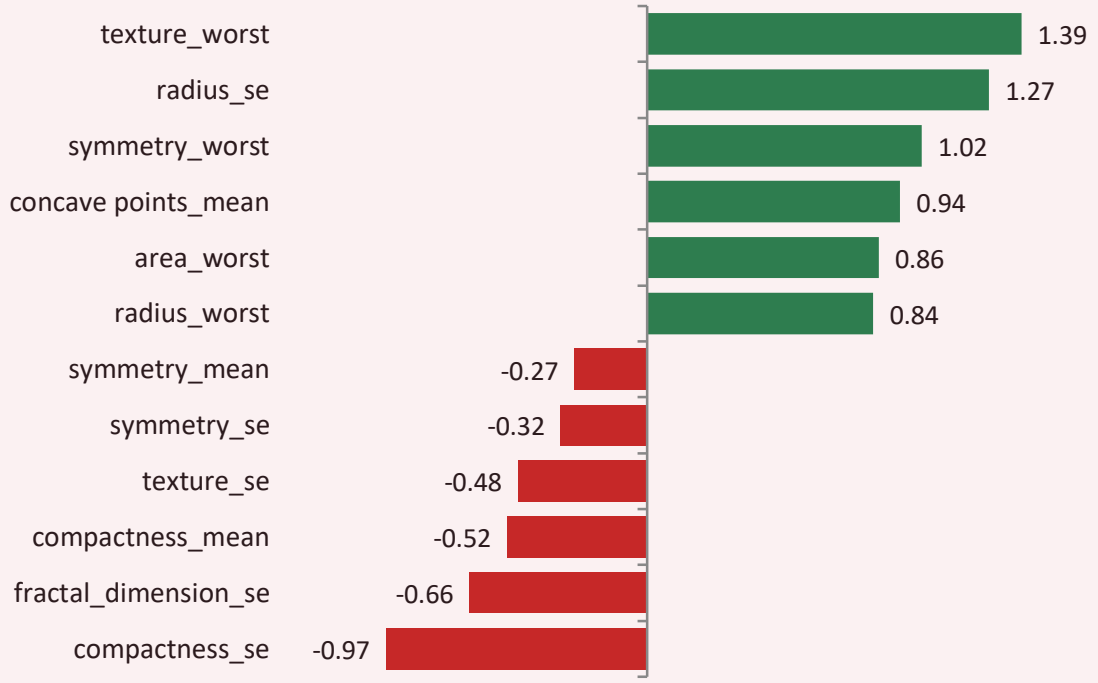
96.50% Accuracy	**98%** Precision
92.45% Recall	**95.15%** F1-Score

Confusion matrix (Testing Set)

Actual \ Predicted	Benign (0)	Malignant (1)
Benign (0)	**89**	**1**
Malignant (1)	**4**	**49**

Feature coefficients: impact on model predictions

Six strongest positive and six strongest negative coefficients (of 30 features)



Key takeaway: The model achieves high accuracy and effectively distinguishes between benign and malignant cases.

Learnings

1 What Worked, What Didn't

Used Jupyter Notebook in addition with Python code. Had problems with executing the code via a single python file, so made an ipynb file instead for executing the code block by block.

2 Who Did What

Yadidya Roy · PPT Lead
Designed and prepared the presentation and coordinated its overall structure.

Abhimanyu Sharma · Execution Lead
Oversaw the implementation and execution of the project.

Shreya Sandip · Video Lead
Managed video preparation and presentation visuals.

Anadi Krishn Agnihotri · Documentation Lead
Handled project documentation and written materials.

All team members contributed collaboratively towards completing the project.

3 AI-Use Disclosure

- AI tools were used at different stages: understanding logistic regression and statistical concepts, debugging Python code, preparing documentation and improving visualizations and the README.
- Gemini was used to create the code for the feature importance (coefficients) graph in logistics_model.py.
- The dataset is based on the Wisconsin Diagnostic Breast Cancer dataset. Model predictions and evaluation results come from running the implemented machine learning pipeline.